

Lipschitz monotonicity alone does not ensure FBF convergence

Disproof of conjecture 00000008832

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Both versions of the conjecture assert unconditional convergence of forward-backward-forward splitting under joint Lipschitz monotonicity. Neither states existence of a zero of the inclusion. We give an actual FBF iteration with monotone Lipschitz operators and a step satisfying the usual strict Lipschitz step bound, but with no weak or strong limit. The example has no zero; it does not contradict the standard convergence theorem that assumes a solution exists.

Actual operators and their hypotheses. Work in the real Hilbert space $\mathbb{R}$, and set

$$A(x) = \{0\}, \quad B(x) = 1 \quad (x \in \mathbb{R}).$$

Both graphs are maximal monotone. To prove this directly, for any $c \in \mathbb{R}$ the graph $G_c = \{(x, c) : x \in \mathbb{R}\}$ is monotone, because its monotonicity inner product is zero. If a point (p, q) can belong to a monotone graph containing G_c , comparison with $(p - 1, c)$ and $(p + 1, c)$ gives

$$q - c \geq 0, \quad -(q - c) \geq 0.$$

Thus $q = c$, proving maximality against every monotone graph extension.

The single-valued versions of A and B are both constant and hence zero-Lipschitz. In particular each is 1-Lipschitz, so the usual positive Lipschitz upper bound $L = 1$ is valid.

Genuine resolvent and three stages. For any $\gamma > 0$, the resolvent equation for A is

$$x = u + \gamma a, \quad a \in A(u) = \{0\}.$$

It holds exactly when $u = x$. Thus the resolvent is the actual unique map $J_{\gamma A} = \text{id}$. The standard FBF stages are

$$v = x - \gamma B(x), \quad y = J_{\gamma A}(v), \quad x^+ = y - \gamma(B(y) - B(x)).$$

Here $v = x - \gamma$, $y = v$, and $B(y) - B(x) = 0$. Consequently

$$x^+ = x - \gamma.$$

The backward stage is well-defined everywhere and solves its resolvent equation exactly; no approximate certificates are assumed.

All iterates and absence of a limit. Choose $\gamma = 1/2$. It satisfies $0 < \gamma < 1/L$ for $L = 1$, so the example does not use a violation of the usual Lipschitz bound. For any initial point $x_0 \in \mathbb{R}$, induction gives

$$x_n = x_0 - n/2 \quad (n \geq 0).$$

In particular the trajectory diverges to minus infinity. There is no finite strong limit. One direct argument, used formally, is that convergence of x_n to a would imply convergence of the shifted sequence x_{n+1} to a and hence of its difference to zero. But

$$x_{n+1} - x_n = -1/2$$

is a nonzero constant sequence, which cannot converge to zero.

There is also no weak limit: the identity on $\mathbb{R}$ is an actual continuous linear functional, so weak convergence would imply the strong convergence just excluded. The same argument works for every nonzero fixed step, not just $1/2$.

The missing hypothesis. The actual inclusion is

$$0 \in A(x) + B(x) = \{1\},$$

which has no solution. Joint monotonicity and Lipschitz continuity do not imply nonemptiness of the zero set. Thus the source's unqualified convergence assertion is false; a valid standard FBF result needs a solution-existence hypothesis in addition to its operator and step assumptions. This disproof makes no separate assertion about adaptive backtracking or the detailed joint domain of unequal step sizes.

Formal correspondence. The Lean project defines actual graphs as subsets of $\mathbb{R}^2$ and monotonicity using the real Hilbert inner product. `constant_maximal` proves inclusion-maximality against every monotone graph extension. `joint_lipschitz` proves actual Mathlib `LipschitzWith` bounds for both operators.

`actual_resolvent` proves the resolvent equation and its unique solution. The forward, backward and final forward stages are defined explicitly; `genuine_backward_stage` verifies the exact equation. `actual_step` and `actual_iterates` prove the update and all iterates for every starting point and every step. `no_strong_limit` proves nonconvergence, and `no_weak_limit` uses the full standard functional definition quantified over all continuous linear functionals. `no_zero` checks the actual inclusion. The final theorem combines all hypotheses and the absence of any weak limit for the safe half-step.

Reproduction. Use Lean 4.19.0 and public Mathlib revision

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Run `lake build` inside `lean/`. Axiom audits are printed, and the validation notes record the completed build and PDF inspection.